

BARBARA FAGUNDES, Ph.D.

Postdoctoral Researcher | Engineering & Computing Education (K–12 to Higher Ed)

Computational Thinking | AI in Education

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PROFESSIONAL SUMMARY

Postdoctoral researcher in Engineering and Computing Education with experience spanning K–12 and higher education, with expertise in qualitative and mixed-methods research on student learning in STEM contexts. Work focuses on computational thinking and the integration of generative AI in educational settings, including work on student engagement with AI in undergraduate courses. Brings experience translating research into instructional practice through faculty collaboration, student–faculty partnerships, and initiatives supporting teaching and learning innovation across educational levels.

RESEARCH INTERESTS

- AI in Education and Generative AI in Learning Contexts
- Computational Thinking across K–12 and Higher Education
- Student Learning, Motivation, and Engagement in STEM
- Translational Research for Teaching Practice and Educational Innovation

EDUCATION

Purdue University

West Lafayette, IN

Ph.D., Engineering Education

Winter 2024

Jackson State University

Jackson, MS

M.S., Computer Science

Spring 2017

Universidade Sant'anna

São Paulo, Brazil

B. A., Early Childhood Education

Minors: K-5 education, policy, and school administration

Summer 2010

RESEARCH EXPERIENCE

Purdue University

Postdoctoral Researcher

Center for Instructional Excellence (CIE)

West Lafayette, IN

Fall 2024 – Present

- Lead qualitative research on undergraduate students' use of generative AI in computer science and engineering courses
- Designed and facilitated focus groups and interviews; analyzed qualitative and survey data
- Co-authored manuscripts and presentations translating findings into evidence-based recommendations for teaching and learning
- Co-lead the Student Pedagogy Advocates (SPA) program, facilitating student–faculty partnerships and mentoring undergraduate partners in research–practice initiatives
- Collaborate with faculty and institutional teaching initiatives to support evidence-based instructional practices and the integration of emerging technologies, including AI
- Contributed to research on learner motivation and autonomy using Self-Determination Theory

Purdue University

Graduate Research Assistant at the NSF RECT project

(Rethinking Circle Time: Integrating Computational Thinking into K-2 Literacy)

West Lafayette, IN

Spring 2022 – Fall 2023

- Developed programming lessons integrating computational thinking into K–2 literacy curriculum
- Created instructional video tutorials including scripting, filming, and editing
- Conducted classroom-based video data collection and qualitative analysis in K-2 classroom settings
- Performed literature reviews in K–12 STEM and computational thinking education, with a focus on early childhood (K–2)
- Co-authored peer-reviewed journal and conference papers
- Designed and administered Qualtrics surveys to assess teachers' computational thinking implementation
- Mentored junior PhD students and supported collaborative research activities

Purdue University

Graduate Research Assistant

(Epistemic Research Laboratory)

West Lafayette, IN

Fall 2018 – Fall 2019

- Coordinated classroom-based video data collection in K–12 engineering settings
- Managed research logistics to ensure timely and compliant data collection
- Supported implementation of engineering design curriculum in K–12 classrooms
- Co-authored peer-reviewed conference paper on engineering design decision processes

Purdue University
Data Research Manager
(Research Institute for Pre-College Engineering - INSPIRE)

West Lafayette, IN
Fall 2017 – Fall 2018

- Managed research data and classroom-based video collection for grades 6–8
- Coordinated logistics for large-scale pre-college engineering research initiatives
- Conducted computational thinking activities for K–12 students
- Collaborated with faculty and graduate researchers to streamline research

TEACHING EXPERIENCE

Purdue University, School of Engineering Education
Teaching Assistant, E-learning development (Graduate Course)

West Lafayette, IN
Spring 2024 – Summer 2024

- Co-developed the graduate course Teaching Engineering Online
- Applied instructional design principles to structure modules and align learning outcomes
- Designed accessible and interactive course materials
- Collaborated with faculty collaborators to integrate evidence-based teaching practices

Purdue University
Teaching Assistant (Instructional Design)
CISTAR NSF Engineering Research Center

West Lafayette, IN
Summer 2021

- Designed e-learning training modules for middle school teachers within NSF-funded engineering initiatives
- Collaborated with faculty and project partners to align instructional materials with learner needs and curriculum objectives
- Authored video scripts and produced instructional lessons and tutorials
- Utilized Camtasia and Adobe Premiere Pro to develop high-quality digital learning materials
- Completed instructional development within established timelines and project guidelines

Purdue University – CISTAR NSF Engineering Research Center
Teaching Assistant (Curriculum Development)

West Lafayette, IN
Summer 2020

- Collaborated with interdisciplinary research team members to develop a middle school chemical engineering curriculum
- Identified student learning needs through communication with faculty and iterative feedback cycles
- Established instructional goals and aligned activities with engineering learning objectives
- Participated in weekly progress meetings to refine curriculum design and implementation

Purdue University – School of Engineering Education
Teaching Assistant, First-year Engineering Courses

West Lafayette, IN
Fall 2019 – Fall 2021

- Supported instruction in two required First-Year Engineering courses serving ~240 students per semester
- Mentored student teams through semester-long engineering design projects
- Guided teams through problem scoping, stakeholder analysis, criteria and constraints development, and iterative design refinement
- Supported students in using Excel and MATLAB to evaluate design decisions
- Provided feedback on technical reports, visual communication, and oral presentations
- Led select class sessions as guest instructor
- Coordinated undergraduate TAs and graders, aligning grading practices and rubrics
- Held regular office hours supporting programming assignments, design challenges, and project development

Jackson State University
Tutor of Computer Science Subjects
(Digital Logic, Physics, and Computer Science Courses)

Jackson, MS
Fall 2015 – Spring 2016

- Provided academic support to undergraduate students in Digital Logic, Physics, and Computer Science courses
- Assisted students with programming assignments and lab activities
- Supported students in troubleshooting coding and logic-based problems

Pre-Kindergarten and Elementary School Teacher
Teacher in private and public schools
(Literacy, Mathematics, Geography, History, Science)

São Paulo, SP (Brazil)
Fall 2008 – Summer 2013

- Designed and delivered interdisciplinary lessons in literacy, mathematics, science, geography, and history
- Implemented project-based learning activities to support early problem-solving and collaboration skills
- Differentiated instruction for students with diverse learning needs and backgrounds
- Conducted monthly progress assessments and prepared detailed reports for parents and school administration
- Led parent meetings and participated in faculty coordination meetings

JOURNAL AND PEER-REVIEWED CONFERENCE PUBLICATIONS

Dickey, E., Bejarano, A., Kuperus, R., & **Fagundes, B.** (2025). *Evaluating the AI-Lab intervention: Impact on student perception and use of generative AI in early undergraduate computer science courses*. *arXiv preprint* arXiv:2505.00100.

Fagundes, B., & Moore, T. J. (2025, June). *Comparing computational thinking learning and engagement in first-grade boys and girls: A study of algorithm design and debugging (WIP)*. In *Proceedings of the ASEE Annual Conference & Exposition*, Pre-College Engineering Education Division, Montréal, Canada. (This paper draws from Dr. Fagundes's doctoral dissertation research on engagement and computational thinking in early childhood education.)

Fagundes, B., Guberman, D., Smart, K., & Holder, K. (2025, June). *Student Pedagogy Advocates: Enhancing teaching and learning through student-faculty partnerships (WIP)*. In *Proceedings of the ASEE Annual Conference & Exposition*, Faculty Development Division, Montréal, Canada. <https://doi.org/10.18260/1-2--55986> (This work builds on Dr. Fagundes's current postdoctoral research on faculty development and autonomy-supportive teaching practices.)

Fagundes, B., & Bhide, N., & Moore, T. J., & Tank, K. M. (2021, July), *Computational Thinking in First Grade Students Using a Computational Device (WIP)* Paper presented at 2021 ASEE Virtual Annual Conference Content Access, Virtual Conference. 10.18260/1-2—36825

Fagundes, B., & Ehsan, H., & Moore, T. J., & Tank, K. M., & Cardella, M. E. (2020, June), *First-graders' Computational Thinking in Informal Learning Settings (WIP)*. Paper presented at 2020 ASEE Virtual Annual Conference Content Access, Virtual Online. 10.18260/1-2--35541.

Johnston, A. C., & Lopez-Parra, R. D., & Tank, K. M., & Moore, T. J., & **Fagundes, B.** (2019, June), *Design Decision Processes of First Grade Students during an Engineering Design-based STEM Unit (Fundamental)* Paper presented at 2019 ASEE Annual Conference & Exposition, Tampa, Florida. 10.18260/1-2—32599

Yeter, I. H., & Ryneerson, A. M., & Ehsan, H., & Rehmat, A. P., & Dasgupta, A., & **Fagundes, B.**, & Menekse, M., & Cardella, M. E. (2019, June), *Design and Implementation of Data Collection in a Large-Scale, Multi-Year Pre-College Engineering Study: A Retrospective*. Paper presented at 2019 ASEE Annual Conference & Exposition, Tampa, Florida. 10.18260/1-2—32596

PRESENTATIONS, WORKSHOPS, AND POSTERS

Tank, K. M., Ottenbreit-Leftwich, A., Moore, T. J., Yang, S., Wafula, Z., Kim, J., **Fagundes, B.**, & Chu, L. (2024). Investigating Sequencing as a Means to Computational Thinking in Young Children. *International Journal of Computer Science Education in Schools*, 6(3), 67–77. <https://doi.org/10.21585/ijcses.v6i3.192>

Moore, T., Tank, K., & **Fagundes, B.** (2023). *Task Interviews as a Research Tool: Cases from K–2 Computational Thinking*. Workshop presented at the Pre-College Engineering Education Division (PCEE) at the 2023 ASEE Annual Conference & Exposition, Baltimore, MD.

Ottenbreit-Leftwich, A., Moore, T. J., Tank, K. M., Kim, J., **Fagundes, B.**, Chu, L., & Wafula, Z. (2023). *Rethinking Circle Time: Development of K-2 CT Literacy Integrated Curriculum* [Poster presentation]. SIGCSE 2023: 54th ACM Technical Symposium on Computer Science Education, Toronto, Ontario, Canada. <https://doi.org/10.1145/3545947.3576338>. (pp. 1394-1394).

Fagundes, B., Bhide, N., Moore, T. J., Drummond Oaks, M., & Godwin, A. (2022, June). *Online professional development aid for teaching an engineering design-based curriculum in 8th grade (Resource Exchange)*. Work showcased at the Exhibit Hall in 2022 ASEE Annual Conference & Exposition, Minneapolis, MN.

Lopez-Parra, R. D., **Fagundes, B.**, Wallace, D., Bhide, N., Moore, T. J., Drummond Oaks, M., & Godwin, A. (2022, June). *Engineering design-based curriculum for teaching 8th grade students renewable energy (Resource Exchange)*. Work showcased at the Exhibit Hall in 2022 ASEE Annual Conference & Exposition, Minneapolis, MN.

Fagundes, B., & Bhide, N. (2021, July). *Computational Thinking in First-Grade Students Using a Computational Device*. Virtual Poster presentation at 2021 ASEE Annual Conference & Exposition. Online.

Fagundes, B., & Cardella, M. (2018, December). *Exploring Engineering Behaviors of Young Children Interacting with a Parent*. ENE Explorer Poster Session Presentation at Engineering Education Researcher Seminar. Purdue University in West Lafayette, IN.

HONORS AND AWARDS

- **Outstanding Service Scholarship Award**, Purdue University College of Engineering, 2024

Recognized by the College of Engineering for outstanding service and contributions to the Purdue community.

- **Purdue Society of Women Engineers (SWE)** – WE22 National Conference

Selected and sponsored by Purdue SWE to represent the university at the 2022 Women in Engineering National Conference.

- **Grace Hopper Celebration (GHC)** Student Scholar, Orlando, FL, 2019

Selected as a GHC Scholar to attend the Grace Hopper Celebration, the world's largest conference for women in computing.

- **Diversity Scholarship**, Jackson State University, 2016

Awarded a merit-based scholarship supporting international students from underrepresented backgrounds.

- **Presidential Scholarship for Minority Students (PROUNI – University for All Program),**

Universidade Sant’Anna, São Paulo, Brazil, 2007

Awarded a full-tuition federal scholarship for undergraduate studies through Brazil’s national higher education inclusion program, based on academic excellence and performance on the National High School Exam (ENEM).

ACADEMIC AND PROFESSIONAL SERVICE

- **Reviewer, Journal of Pre-College Engineering Education Research (J-PEER)** Fall 2025–Present
(peer-reviewed journal in engineering education)

- **Brazilian Student Association (BRASA Global)** – Student Brand Ambassador

Summer 2022 – Summer 2023

Volunteer speaker giving virtual presentations at Brazilian universities to promote international scholarships, mentorship programs, and educational opportunities for low-income students. Collaborated with other BRASA ambassadors worldwide to organize crowdfunding initiatives supporting scholarship programs for underrepresented students.

- **Women in Engineering Program (WIEP)** – Pre-College Engineering Outreach Team Member, Purdue University - *Summer 2022 – Summer 2023*

Mentored pre-college girls in engineering through on- and off-campus activities. Collaborated with a team of women engineering students to promote early engagement and interest in STEM fields.

- **Purdue Graduate Student Government (PGSG)** – Senior Senator

Fall 2022 – Spring 2023

Represented the Engineering Education Graduate Student Association (ENEGSA) in PGSG Senate meetings. Contributed to discussions, resolutions, and policy proposals affecting graduate and professional students. Reported outcomes to ENEGSA members to support transparency and shared governance.

- **Grant Review and Allocation Committee (GRAC)** – Reviewer, PGSG

Fall 2022

Reviewed and evaluated Purdue students’ applications for professional development and community-based grants. Provided feedback and funding recommendations based on merit and alignment with program goals.

- **American Society for Engineering Education (ASEE)** – Reviewer

Fall 2018 – Present

Evaluate annual ASEE conference submissions for rigor, clarity, and contribution to the engineering

education community. Provide detailed feedback to authors and editorial recommendations on acceptance and revision.

- **Engineering Education Graduate Student Association (ENEGSA)** – Member, ESL & Multicultural Committee, Purdue University, 2020

Co-created and hosted “*We, the Aliens*” podcast, discussing the experiences of international students in U.S. higher education. Organized workshops and collaborated with Purdue’s Latino Cultural Center to foster intercultural dialogue.

- **Summer Undergraduate Research Fellowship (SURF) e-Symposium** – Judge
2020

Served as a judge for undergraduate research poster presentations in collaboration with the Engineering Undergraduate Research Office (EURO). Evaluated and provided feedback on research quality and presentation effectiveness during the first fully virtual SURF Symposium.

- **School of Engineering Education (ENE), Purdue University** – Academic Mentor
2020

Mentored Ph.D. students in the School of Engineering Education, offering one-on-one guidance on research design, coursework, and teaching practices during the COVID-19 transition to online learning.

PROFESSIONAL CERTIFICATION AND MEMBERSHIPS

- Member American Society for Engineering Education (ASEE)
- Society of Women in Engineering (SWE)

LANGUAGES AND SOFTWARE

Languages: Portuguese (Native), English (Fluent), Spanish (Basic)

Technical and Programming Tools: MATLAB, JavaScript, Python, Java, C++, PHP, SQL.

Digital and Web Tools: HTML, CSS, WordPress, MS Office, LMS tools, Camtasia, Articulate 360, Adobe Premiere Pro, Adobe InDesign.